

Neural Networks Assessment Instructions

University of Sussex, 2026, A2

Overview

Objective

In this assessment, you will be asked to carry out three tasks. The first task involves building and training a simple CNN-based classifier for the Fashion MNIST dataset, and demonstrating the effectiveness of a technique to improve the performance of your classifier on the test dataset. The second task involves building and training an unconditional GAN to generate images that look like those in the Fashion MNIST dataset, and using the discriminator to investigate the quality of the generated images. The third task involves augmenting the GAN to improve the generated images in some way. You will write a report up to 2500 words to describe and explain your methods and results. You will write all of your code and the report in a single Jupyter Notebook (XXXXXX.ipynb), which has been provided in this assessment package.

How to submit

Please submit your .ipynb file online through the Canvas assessment submission point. Only this file will be marked, so please do not submit multiple files or a zipped file that contains multiple files. I recommend you use university computers to complete this work, as they are well equipped with all the tools and hardware you need. You can use VScode to edit the ipynb file. Google colab is also an option, but free access can be very limited and unreliable. Rename the Jupyter Notebook filename, XXXXXX.ipynb, by replacing XXXXXX with your 6-digit candidate number.

Grading

Your assessment will be marked out of 100% according to the marking criteria provided at the end of this document. Grades will be based on how many tasks were completed, the quality with which those tasks were performed and results presented, and the level of research demonstrated by going beyond the taught material by drawing upon work from academic publications.

Important notes

- You must use PyTorch to build, train, and test the NNs for this assessment. You are permitted to use any of the functions available in PyTorch, e.g. to import and process data, to define NN layers and optimisers, etc., but you are not permitted to load pre-trained NN models that have been built by others.
- All figures in your report must be generated as output from a code cell. Imported images will not be graded, even if you first created them as output from a code cell, saved them, and then imported them.
- Make sure that every figure has a corresponding caption that explains the details of what that figure is showing. Make sure that figures have labelled axes. Pay attention to the clarity of your figures. Ensure that the scales and limits of axes are consistent if you are comparing data between figures, and avoid overcrowded figures.

Task summary

Task 1

Build and train a 5 layer CNN to perform classification on the Fashion MNIST dataset. The CNN must take 28x28 greyscale inputs, and provide a 10-dimensional output for multiclass classification. Your report should describe the details of your model in sufficient detail for someone to replicate it without access to your code. This includes all necessary architectural details as well as optimisation details. Good performance on this dataset for a small model like this one would be above 90% accuracy on the test data. You should investigate a range of learning rates to find the one that results in the best performance. Explain and demonstrate how any other features of your model/optimisation also help to improve performance (e.g. by comparing accuracy with/without that feature).

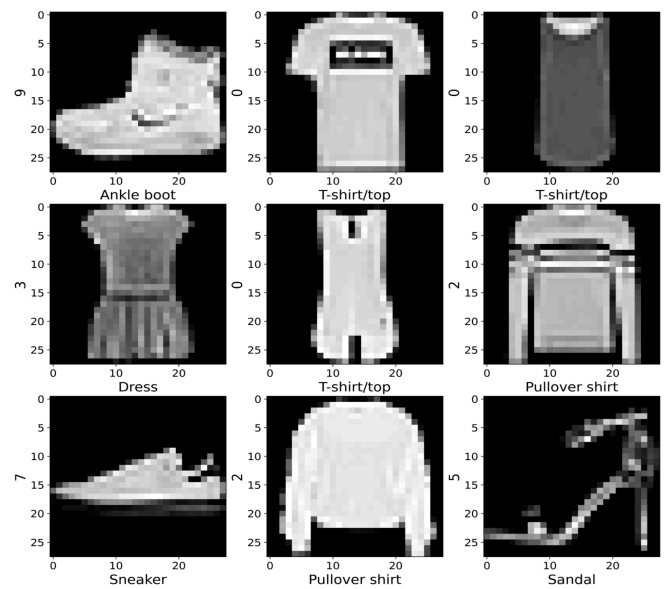


Figure 1. Fashion MNIST dataset

Task 2

Build and train a Generative Adversarial Network in order to generate realistic images within the Fashion MNIST dataset. The input to the generator should be a 32-dimensional random noise vector. Use the minimax loss function to train the Discriminator and Generator. Your report should describe the details of your model in sufficient detail for someone to replicate it without access to your code. This includes all necessary architectural details as well as optimisation details. Use the classifier you built in Task 1 to classify the generated images at various points throughout training. Use the classifier to assess the quality of your GAN, for example in terms of how the classifier probabilities are distributed amongst classes per sample (relates to how realistic the generated image appears), and how often any given class is attributed the highest probability (relates to the diversity of the generated images, which can be shown, for example, in a histogram).

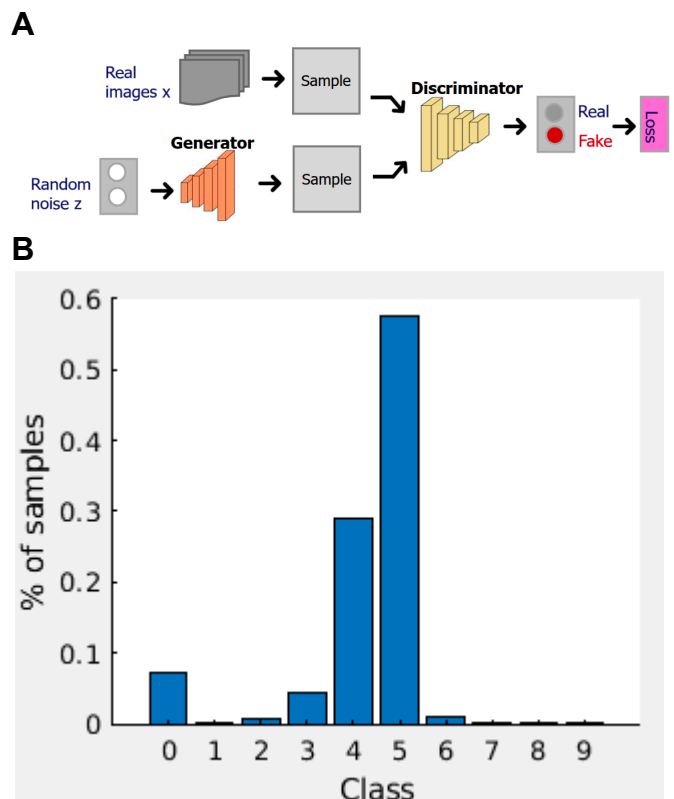


Figure 2. **A)** Illustration of a Generative Adversarial Network, including the Generator, Discriminator, and the real/generated data that is input to the Discriminator. **B)** Example histogram showing the relative percentage of generated samples that are assigned to each class by a classifier.

Task 3

Attempt to improve the quality of your GAN by changing a single feature of it. For example, you might update the loss function, or use a conditional GAN rather than an unconditional GAN. Use your classifier from Task 1 to comment on whether your modification improves the quality of generated images, the diversity of generated images, or both.

Further details

You are given space with the .ipynb file to write code in code cells and to write the text of your report in markdown cells for all 3 tasks. **You must not add or remove Markdown Cells, but you may add code cells, for example to produce different figures.** When writing responses in markdown cells, you must use the cells with text in italics that says *“Replace this text with your report for Task X”*.

Useful tips to format markdown cells can be found in the PDF, **notebook_markdown_cheatsheet.pdf**.

Your submission must include only the .ipynb file. If you upload **any additional Python files, text documents, or images, they will be ignored**. You must rename the notebook by replacing XXXXXX in the filename with your 6-digit candidate number.

Report Figures

All figures for your report must be generated as output from code cells. Therefore, make sure those figures are still visible when you submit your assessment. Any figures that present results, and that have been loaded into the notebook as an image, will not be marked. **I will not run any of your code cells to produce the output.**

Guidance for how to produce figures and how to use Latex markup (e.g. for equations) within the Notebook is provided in the separate Notebook, **useful_code_for_figures_equatons.ipynb**.

Important: every figure should be accompanied by a caption that describes what is being shown in the figure. Each caption should first provide the figure name, e.g. “Figure 1”. **Ensure you refer to figures by their correct name in the written report.**

Word count

Your written report in Markdown cells must not exceed a word limit of 2500 words in total, as determined by the Python script, `count_jupyter_nb_words.py`, that is provided with this assessment. An example of how to use `count_jupyter_nb_words.py` is provided as preamble comments within the file. When writing your responses, remember to delete the text *“Replace this text with your response”*, otherwise it will contribute to your word count. None of the other text, either in code cells, or in markdown cells that describe each subtask, will contribute to the word count. The script outputs the number of words you have added to the document. If you exceed the word count, the script will also output a short string, starting from where the word count is exceeded, so that you can locate where that happens. You are allowed to over this limit by up to 250 words, which will incur a 2% penalty. **Any part of the Markdown cells that exceeds 2750 words will not be marked.**

Code

All code to build, run, and analyse your models, as well as to plot results, must be provided in code cells

within the XXXXXX.ipynb. If you need to create new code cells for a given task, you may do so immediately after the provided code cell. **Important: the person marking your report will not run any of your code.** Therefore, all figures must be generated and visible in the Notebook when you submit to Canvas, or you will miss available marks. In order for those marking your report to determine whether or not you have correctly completed the experiments, they may need to look at your code. To help us make sense of your code, and therefore to help us improve our feedback to you, please **provide plenty of comprehensible comments**, including general overviews for each section of code, and shorter comments every few lines of code (if not on every line). You will receive some marks for good code and good comments.

Other Important points

You are not expected to produce the perfect NN models with competitive performance. What we are looking for is your scientific scrutiny of the model and the techniques you have employed.

Your work will be assessed by humans who get tired/hungry/frustrated/intrigued/curious. While we make every effort to assess your work fairly, those humans are not perfect. Help them to assess your work, and to realise how talented you really are, by making your presentation clear and concise. Check that references to figures are consistent, check for spelling, grammar, and consistent terminology. Every little helps.

Use of generative AI

You are only permitted to use generative AI as a tool to aid your learning experience, and for debugging code. You are not permitted to use generative AI to write the code or the written report for you. If you have any questions about what is and is not permitted use of generative AI, please email me for clarification, james.bennett@sussex.ac.uk.

I have given the tasks and assessment details (as well as paraphrased versions) to different generative AI tools (e.g. chatGPT, Gemini, Copilot etc.) and generated numerous responses from it. I therefore have a good idea of the kind of content it will produce for this assignment. Be very cautious about what these tools provide. Remember: **these are generative models**, which means that **they have been trained to give responses that prioritise appearing plausible over being factually correct**. It is not difficult to get generative AI to generate factually incorrect responses about technical details. In previous assessments, some submissions clearly use generative AI, as they contain remarkably similar responses that miss out on distinct details that were needed to score marks. Generative AI also tends to be overly verbose, which can risk pushing your submission over the word count and consequently missing out on marks.

Marking criteria

Your final mark will be determined by a combination of factors: the number of Tasks you complete, the content you provide for each Task, and the quality with which that content is presented:

$$final_mark = content_grade \times quality_grade \times upper_limit(Tasks_completed)$$

The word limit is 2500 words. This applies to the content of Markdown cells only, and excludes code, Figure captions, LaTeX equations, and the Bibliography. Exceeding the word count by up to 250 words will incur a 2% penalty. Thus, **2750 words acts as a hard upper limit: any content that goes beyond 2750 words will not be marked.**

Completion of Tasks

Completing Task 1 only will result in marks being capped at 50%. Completing Tasks 1 and 2 only will result in marks being capped at 70%. Only by completing all tasks can marks up to 100% be achieved. Important: completing all 3 Tasks does not guarantee a mark greater than 70%, as your final mark also depends on the quality of your work.

Task Content

For each Task, your report is expected to show evidence of the following:

- Concise and detailed methodological descriptions that would allow the reader to implement your model and optimisation without having access to your code.
- Concise discussion of the advantages and disadvantages of different features/techniques used within your model/optimisation. These should be statements about *how/why* the technique improves performance.
- Clear and concise explanations of the experiments you performed, how they were performed, the results, and the conclusions you drew from the results.
- An objective and honest discussion of how your model performed, what its limitations/weaknesses are, how those limitations/weaknesses could be mitigated and improved upon. The discussion will be well-reasoned, with explanations for *how/why* the limitations/weaknesses/improvements affect performance.
- References to published academic papers are provided, where relevant. In particular, Task 3 provides a good opportunity to go beyond taught material, so references to the techniques you use are crucial there. References should be cited in-line in the text immediately after the relevant content — e.g. (Bloggs et al., 2021) — and the full reference should be given in the provided Markdown cell at the end of the .ipynb file, e.g. “Bloggs, Blaggs & Bleggs (2021), Title of the article, Journal/conference name, DOI: <https://doi.org/10.XXXX/YYYY...>”
- Figures should present all of the data/results necessary to support your written description of results and the conclusions you draw from them.
- Figure axes will be appropriately scaled and use appropriate limits so that the data/results can be correctly interpreted, and so that comparisons can be made between multiple plots where relevant. Axes will be labelled with meaningful titles, and have appropriate tickmarks.
- Figures will have captions that provide concise yet thorough descriptions of the data being shown in the figure’s plots. Figures will be named, e.g. “Figure 4”, as will different plots within a figure, e.g. “A”, “B”, etc. The written report will refer to figures in the text, where relevant, e.g. “*the generalisation gap decreased when we included method X (Figure 4A), resulting in an accuracy of 96% on the test set (Figure 4B)*”.
- Code will be well-commented to enable the reader to easily understand what the code is achieving. This will entail long comments to describe the purpose of a large block of code, and short in-line comments to describe the purpose of individual lines of code. The written report will refer to code, where relevant, by stating the cell and line number where the relevant code begins, e.g. “*layers 1-4 used 3x3 convolutional filters with a stride of 1 and padding of 1 (cell 5, line 30)*”.

Content Quality

Finally, the quality of the work you submit will be assessed according to the following criteria:

- **70%-100%:**
 - Shows excellent understanding of the methods used for developing neural network solutions, supported by evidence that the student has extrapolated from what was taught, through extra study or creative thought. Work at the top end of this range is of exceptional quality. The report will be excellently structured, with proper references to academic publications and a proper discussion of relevant, academic work. The report will be neatly presented, interesting and clear, with a disinterested, objective critique of what did and did not work well with the approach taken, and a well-reasoned discussion about how the work could be usefully extended. It will be concise yet detailed, making use of strong, scientifically grounded explanations for the methods chosen, and for the results that are observed.
- **60-69%:**
 - The work will be very competent in all respects, evidencing substantially correct and complete knowledge of neural network solutions, though will not go beyond what was taught. The report should be well-structured and presented with proper references to academic publications and some discussion and objective critical evaluation of the approach taken, with a discussion about how the work could be usefully extended. Presentation will generally be of a high standard, with clear and mostly concise written style, making use of scientifically grounded explanations for the methods chosen, and for the results that are observed.
- **50-59%:**
 - The work will be competent in most respects. There may be minor gaps in knowledge, but

the work will show a reasonable understanding of fundamental concepts about neural network solutions. The report will be generally well-structured and presented with some references to academic publications, though these may not always be relevant. The report may lack clarity and depth to its critical discussion, and to the discussion of how the work may be usefully extended. It will provide some explanations for the methods chosen, and for the results that are observed, but they may lack some clarity and correctness.

- **40-49%:**
 - The work will have some significant gaps in knowledge but will show a basic understanding of fundamental concepts of neural network solutions. The report should cover the fundamentals but may not cover some aspects of the work in sufficient detail. The work may not be organised in a clear and logical way, and its presentation may not always be appropriate or useful. There will be minimal critical evaluation or discussion. References may be missing or unsuitable. The report will provide some explanations for the methods chosen, or for the results that are observed, but they may not be correct or well-explained.
- **30-39%:**
 - The work will show inadequate knowledge of the subject. The work is seriously flawed, displaying a major lack of understanding, irrelevance, or incoherence. The report is badly organised and incomplete, possibly containing irrelevant material. The report may be missing some required content.
- **Below 30%:**
 - The work is either not submitted or, if submitted, so seriously flawed that it does not constitute a bona-fide report.